

course outcomes.

CO 1 -

Define and illustrate fundamental set theory concepts, logical connectives and propositional logic

CO 2 -

Analyzed different types of functions and relations including injective, surjective, bijective, reflexive, symmetric, transitive relations.

CO 3 -

Solve problems related to graph theory including identifying shortest path euler and hamiltonian and graph isomorphism.

CO 4 -

evaluate Algebraic structures with one or two binary operations such as groups, rings, fields and perform operations within these structures

C05 -

construct and simplify boolean functions using identities and representations such as disjunctive and conjunctive normal form

SETS -

- A set is a group of objects

Eg, (1) people in class : { Ram, Neel, shyam }
 (2) colors of Rainbow : { violet, Indigo, Blue, Green, yellow, orange, Red }

- Set can contain non-related & elements

Eg, A : { 3, α , red, cycle, 1, 2 }

A set is an ~~or~~ unordered collection of objects called as elements or member of sets.

Properties of sets -

- orders does not matters - we often write in order because it is easier for humans to understand it

A : { 1, 2, 3, 4, 5 }

A : { 3, 4, 5, 1, 2 }

- sets do not have duplicate elements

A : { Ram, shyam, Ram, Neel }

- Set can use set builder notation

A = { x/x is prime no and $x > 2$ }

A : { 3, 5, 7, 13 }

- A set is said to "contain" the various "members" or "elements" that make up the set

A = { 1, 2, 3, 4, 5 } $4 \in A$

• Universal Set

- U is the universal set the set of all elements from which ^{any} set is drawn

Eg (a) $U = \{ \text{set of all natural number} \}$

(2) $A = \{1, 2, 3, 4\}$

$B = \{5, 6, 7\}$

$U = \{1, 2, 3, 4, 5, 6, 7, \dots\}$

• Subset

- if all the elements of set S are also the elements of set T then S is the subset of T

Eg., $S = \{1, 2, 3, 4\}$

$T = \{3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$

$S \subseteq T$

• Empty set

- If the set has zero element is called the empty or null set. $\emptyset = \{ \}$, $A = \{ \}$

• Set Equality

- Two sets are equal if they have the same elements

Eg., $S = \{1, 1, 2, 3, 4\}$

$T = \{1, 3, 4, 2, 1\}$

$S = T$

• Set cardinality

- The cardinality of a set is the number of elements in a set

Eg., $S = \{1, 2, 3, 4\}$

$|S| = 4$

• Power set

- Set of all the subset of that set

Eg., $S = \{0, 1\}$

$= 2^2$

$= \{ \emptyset, \{0\}, \{1\}, \{0, 1\} \}$

Eg., $S = \{a, b, c\}$

$= 2^3$

$= \{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$

• Cartesian Product

- Cartesian product is the direct product of set

Eg., $A = \{1, 2, 3\}$ $B = \{1, 2, 3\}$

$C = \{ (1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), \dots \}$

• Finite and infinite

- A finite set is a set that has a finite number of element

- Eg., R = {violet, indigo, red, yellow, blue, green, orange}
- A set that is not finite is infinite

* Operation on sets

(1) Union operation

For a set A, B the union $A \cup B$ is the set containing all elements that are either in A or in B or in both.

Eg., $A = \{a, b, c\}$

$B = \{d, e, f\}$

$A \cup B = \{a, b, c, d, e, f\}$

(2) Intersection operation

For a sets A, B the intersection $A \cap B$ is the set containing all elements that are simultaneously in A and B.

E.g. (1) $A = \{a, b, c\}$

$B = \{1, 2, 3\}$

$A \cap B = \emptyset$

(2) $A = \{1, 2, 3, 4, 5\}$

$B = \{4, 5, 6\}$

$A \cap B = \{4, 5\}$

(3) Disjoint set

Two sets are said to be disjoint if their intersection is empty.

example: 1) $A = \{1, 2, 3\}$ and $B = \{5, 4, 5\}$

are not disjoint set

2) $A = \{New York, Washington\}$ and $B = \{3, 4\}$ are disjoint set

(4) Complement of set

Complement of a set can be defined as a set that contains the elements present in the universal set but not in set A

example: $U = \{1, 2, 3, 4, 5, 6\}$

$A = \{4, 5, 6\}$

$A' = \{1, 2, 3\}$

(5) Set Difference

For sets A, B the difference of A and B written as $A - B$ is the set of all elements that are in A but not in B

example: $A = \{1, 2, 3, 4, 5, 6\}$

$B = \{2, 3, 5, 7, 9\}$

$A - B = \{1, 4, 6\}$

(6) Symmetric difference

Symmetric difference: [it is denoted by Δ] The symmetric difference includes elements that are in either A or B but not in both.

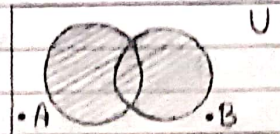
example: $A = \{1, 2, 3\}$ $B = \{3, 4, 5\}$

$A \Delta B = \{1, 2, 4, 5\}$

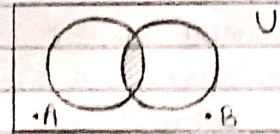
Venn Diagram

A Venn diagram is a pictorial representation of a set in which sets are represented by enclosed area in the plane.

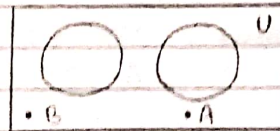
1) Union Venn diagram $[A \cup B]$



2) Intersection Venn diagram $[A \cap B]$



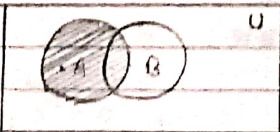
3) Disjoint Venn diagram $[A \text{ and } B \text{ are disjoint}]$



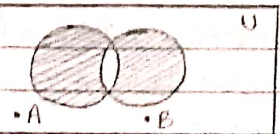
4) Complement Venn diagram (A')



5) Set difference Venn diagram $[A - B]$



6) Symmetric difference Venn diagram $[A \Delta B]$



Que: Find cartesian product of

$$A = \{a, b, c\}$$

$$B = \{2, 3\}$$

→ cartesian product of $A \times B = \{(a, 2), (a, 3), (b, 2), (b, 3), (c, 2), (c, 3)\}$

answer →

Let $A = \{a, b, c\}$ and $B = \{2, 3\}$

Find $A \times B$

We know that the definition of cartesian product

$$A \times B = \{(x, y) : x \in A \text{ and } y \in B\}$$

$$\therefore A \times B = \{(a, 2), (a, 3), (b, 2), (b, 3), (c, 2), (c, 3)\}$$

$$n(A \times B) = 6$$

$$n(A) = 3 \text{ and } n(B) = 2$$

$$n(A \times B) = n(A) \cdot n(B) = 3 \times 2$$

$$\therefore n(A \times B) = 6$$

Que: IF $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$$A = \{2, 4, 6, 8\}$$

$$B = \{1, 3, 5, 7, 9\}$$

$$C = \{1, 2, 3, 4, 5, 6\}$$

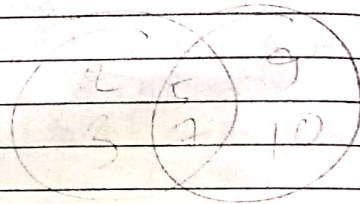
$$D = \{5, 6, 7, 8, 9, 10\}$$

Find $A \cup B, B \cup D, A \cap B,$

$$C \cap D, A - B, D - A$$

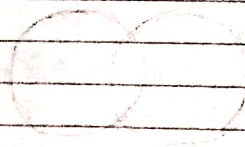
→

- $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $B \cup D = \{1, 3, 5, 6, 7, 8, 9, 10\}$
- $A \cap B = \{3\}$
- $C \cap D = \{5, 6\}$
- $A - B = \{2, 4, 6, 8\}$
- $D - A = \{5, 7, 9, 10\}$



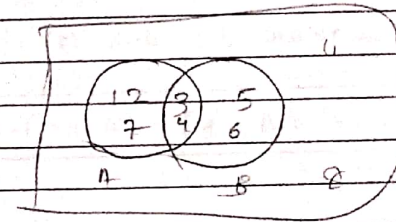
Que: If $A = \{2, 3, 5, 7\}$ and $B = \{5, 7, 9, 10\}$ represents A and B by venn diagram

→ The required venn diagram for the given set is



Que: From the adjoining diagram, find the following

- 1) U
- 2) $A \cap B$
- 3) $A \cup B$
- 4) $(A \cup B)'$
- 5) $(A \cap B)'$



→ $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

→ $A \cap B = \{3\}$

→ $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$

→ $(A \cup B)' = \{8\}$

→ $(A \cap B)' = \{1, 2, 5, 6, 7, 8\}$

Que: If $A = \{2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{2, 4, 6, 8\}$ then verify $A \cap (B - C) = (A \cap B) - (A \cap C)$

$B - C = \{3, 5\}$

$A \cap (B - C) = \{3\} \quad \dots (1)$

$$A \cap B = \{3, 4\}$$

$$A \cap C = \{2, 4\}$$

$$(A \cap B) - (A \cap C) = \{3\} \quad \text{--- (2)}$$

from equation (1) and (2)

$$\therefore A \cap (B - C) = (A \cap B) - (A \cap C)$$

Ques: If $A = \{4, 5, 8, 12\}$ $B = \{1, 4, 6, 9\}$
 $C = \{1, 2, 4, 7, 8, 10\}$ then find

$$\textcircled{1} A - B \quad \textcircled{2} B - C \quad \textcircled{3} A - C \quad \textcircled{4} A - (B - A)$$

$$\textcircled{5} A - (C - B)$$

Ans:

$$\text{Let } S_1 = \{1, 2, 3\}, S_2 = \{3, 4, 5, 6\} \text{ then}$$

Find

$$1) S_1 \cup S_2 \quad 2) S_1 \cap S_2$$

$$3) S_1 - S_2 \quad 4) S_2 - S_1$$

• Set Identities

- (1) Identity law Ex: $U = \{2, 4, 6, 8, 10, 12, 14\}$
- (1) $A \cap \emptyset = A$ $A = \{4, 6, 8, 10\}$
- (2) $A \cap U = A$ $\emptyset = \{ \}$

(2) Domination law

- (1) $A \cup A = A$
- (2) $A \cap \emptyset = \emptyset$

(3) Complementation law

- (1) $\overline{\overline{A}} = A$

(4) commutative law

- (1) $A \cup B = B \cup A$
- (2) $A \cap B = B \cap A$

(5) Associative law

- (1) $A \cup (B \cap C) = (A \cup B) \cap C$
- (2) $A \cap (B \cup C) = (A \cap B) \cup C$

(6) distributive law

- (1) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- (2) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(7) Demorgan's law

- (1) $\overline{A \cup B} = \overline{A} \cap \overline{B}$
- (2) $\overline{A \cap B} = \overline{A} \cup \overline{B}$

(8) Absorption law

$$(1) A \cup (A \cap B) = A$$

(2) $A \cap (A \cup B) = A$

* Properties of Complement

$$17 \quad (\overline{\overline{A}}) = A$$

$$2) \quad A \cup \bar{A} = U$$

5) $A \cap \bar{A} = \emptyset$

4) $\bar{\phi} = 0$

5) $\dot{U} = \emptyset$

* Properties of universal set

$$\Rightarrow A \cup U = U$$

$$2) \quad A \cap U \neq A$$

* Properties of empty set

$$\hookrightarrow A \cup \emptyset = A$$

2) $A \cap \emptyset = \emptyset$

- Mathematical logic

- It is the way/provision to check that given argument or condition is valid or not
- Statement : statement is a declarative sentence having only truth value either true or false.

Eg., (1) India is my country (True)

(2) $9 + 3 = 12$ (True)

(3) six is less than 3 (false)

(4) all the roses are red in color (false)

Eg., Not a statement

(1) open the door

(2) God bless you.

- Types of statements :

(1) Simple / automatic / primary statements not having any connectives.
eg., (1) India is my country (True)

(2) compound / molecular statement

Statements having connectives

eg., Jack and Jill went up the hill
 ↓
 connectives

* Connectives

- (1) Negation ($\neg$, $\rightarrow$) {Not}
- (2) conjunction ($\wedge$) {and}
- (3) Disjunction ($\vee$) {or}

(1) Negation ($\neg$, $\rightarrow$)

eg., (1) P = India is my country

(2) $\neg P$ = India is not my country.

• $\neg P$ = It is not the case that India is my

~~(2) conjunction ($\wedge$)~~ country.

define:

if the statement formed by introducing the word (not $\neg$) at proper place in the statement or by prefixing the place (it is not the case that)

Truth table:

P	$\neg P$
T	F
F	T

(2) conjunction ($\wedge$)

- The conjunction of two statements P and Q is the statement $P \wedge Q$ which is read as $(P \wedge Q)$ the statement P and Q has the truth value T whenever both have the truth value T. otherwise it has the truth value F.

Ex., P = sun is shining today.

Q = There are 15 chairs in classroom

$P \wedge Q$ = sun is shining today and there are 15 chairs in classroom.

Truth table.

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

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(3) Disjunction ($\vee$)

The disjunction of two statements P and Q is the statement $P \vee Q$ which

Eg., P : I will watch the game on TV.

Q : I will go to the game zone.

$P \vee Q$: I will watch the game on TV or go the game zone.

Truth table:

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Note: Normally in everyday language disjunction is used between two statements which have same kind of relationship relation between them.

* Conditional connective / Single Implication ($\rightarrow$)

If P and Q are the statements then the statement $P \rightarrow Q$ which is read as "If P then Q" or "P implies Q" is called conditional statement.

The statement $P \rightarrow Q$ has the truth value F when Q has the truth value false and P has the truth value T. otherwise it has the truth value False.

Ex: P: Pravin takes Mathematics.

Q: Sanjay takes statistics.

$P \rightarrow Q$: If Pravin takes Mathematics then Sanjay takes statistics.

Truth table:

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

* Bi-conditional / Double implication ($\leftrightarrow$)

For any two propositions P and Q, the statement "P if and only if Q" is called as Bi-conditional ($P \leftrightarrow Q$).

The implication is true when P and Q have same truth values and it is false otherwise.

Truth table:

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Que: Write the following statement in symbolic form:

J: Jerry takes Maths.

K: Ken takes Sociology.

L: Larry will take English.

If Jerry takes maths or Ken takes Sociology then Larry will take English.

Symbolic notation: $J \vee K \rightarrow L$

Truth table:

J	K	L	$J \vee K$	$J \vee K \rightarrow L$
T	T	T	T	T
T	T	F	T	F
T	F	T	T	T
T	F	F	T	F
F	T	T	T	T
F	T	F	T	F
F	F	T	F	T
F	F	F	F	T

que: First frame statement then input it onto symbolic form also form the truth table for given statements.

C: Crop will destroyed.

F: There is a Flood.

$F \rightarrow C$: If there is a flood then crop will destroyed.

Que: Let P, Q and R be the following statements

P: He will buy a new computer.

Q: He will celebrate all night.

R: He will win the lottery.

Express the following English sentences as symbolic expression then construct truth table

Statement: ~~If he will win~~

If he will win the lottery, then he will celebrate all night, and if he doesn't win lottery, then he will not get new computer.

Symbolic expression:

$$R \rightarrow (Q \wedge \neg R) \rightarrow \neg P$$

Truth table:

P	Q	R	$\neg R$	$\neg P$	$Q \wedge \neg R$	$R \rightarrow (Q \wedge \neg R)$	$R \rightarrow (Q \wedge \neg R) \rightarrow \neg P$
T	T	T	F	F	F	F	T
T	F	F	T	F	F	T	F
T	F	T	F	F	F	F	T
T	F	F	T	F	F	T	F
F	T	T	F	T	F	F	T
F	T	F	T	T	F	T	T
F	F	T	F	T	F	F	T
F	F	F	T	T	F	T	T

Ques: Construct the truth table for

1) $(P \rightarrow Q) \wedge (Q \rightarrow P)$

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

2) $\neg(P \wedge Q) \leftrightarrow (\neg P \wedge \neg Q)$

P	Q	$P \wedge Q$	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$	$\neg(P \wedge Q)$	$\neg(P \wedge Q) \leftrightarrow (\neg P \wedge \neg Q)$
T	T	T	F	F	F	F	T
T	F	F	F	T	F	T	F
F	T	F	T	F	F	T	F
F	F	F	T	T	T	T	T

3) $(P \wedge (P \rightarrow Q)) \rightarrow Q$

P	Q	$P \rightarrow Q$	$P \wedge (P \rightarrow Q)$	$(P \wedge (P \rightarrow Q)) \rightarrow Q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

4) $(P \rightarrow Q) \leftrightarrow (\neg P \vee Q)$

P	Q	$P \rightarrow Q$	$\neg P$	$\neg P \vee Q$	$(P \rightarrow Q) \leftrightarrow (\neg P \vee Q)$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

5) $((P \rightarrow Q) \wedge P) \rightarrow Q$

P	Q	$P \rightarrow Q$	$(P \rightarrow Q) \wedge P$	$((P \rightarrow Q) \wedge P) \rightarrow Q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

6) $(P \wedge Q) \rightarrow \neg$

P	Q	$\neg$	$P \wedge Q$	$(P \wedge Q) \rightarrow \neg$
T	T	T	T	T
T	T	F	T	F
T	F	T	F	T
T	F	F	F	T
F	T	T	F	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

double parenthesis = work

Well-Formed Formula For statement calculus

- A Statement formula is an expression which is a string consisting of variables, parenthesis and connective symbol.
- Every string of this formula can said to be a well-formed formula.
- By a recursive definition of statement formula. we formed a formula can be generated by rules.
 - 1) A statement variable standing alone is a well-formed formula.
 - 2) If P is a well-formed formula then $\neg P$ is a well formed formula.
 - 3) If P and Q are well-formed formula then $(P \wedge Q)$, $(P \vee Q)$, $(P \rightarrow Q)$ and $(P \leftrightarrow Q)$ are well-formed formula.
 - 4) A string of symbols containing statement variable, connective and parenthesis is a well-formed formula, if it can be obtain by finitely many application of the rules 1, 2, and 3.

Example: 1) $\neg (P \wedge Q)$, $(P \vee Q) \wedge (P \rightarrow Q)$ are well-formed formula.

2) $(P \rightarrow Q) \rightarrow \vee Q$ is not well-formed formula
Reason: Because we require two variables for disjunction or proper parenthesis.

3) $(\neg P \wedge Q) \rightarrow Q$ is not well formed formula.
Reason: we require one parenthesis, that is missing.

conjunction ($\wedge$) -

$$T \wedge T = T$$

Disjunction ($\vee$) -

$$F \vee F = F$$

single conditional ($\rightarrow$) -

$$T \rightarrow F = F$$

Double conditional ($\leftrightarrow$) -

$$T \leftrightarrow T = T$$

$$F \leftrightarrow F = T$$

contradiction, Contingency.

A statement that is true for all of its propositional variables. Tautology.

Ex: Construct the truth table for $P \vee \neg P$

$P \vee \neg P$		
P	$\neg P$	$P \vee \neg P$
T	F	T
F	T	T

2) Contradiction: A statement that is false for all possible values of its propositional variables is called Contradiction.

Ex: Construct the truth table for $P \wedge \neg P$

$P \wedge \neg P$		
P	$\neg P$	$P \wedge \neg P$
T	F	F
F	T	F

3) Contingency: A propositional functional that is neither tautology nor contradiction is called contingency.

Ex: Construct the truth table for $((P \wedge Q) \leftrightarrow P)$

P	Q	$P \wedge Q$	$(P \wedge Q) \leftrightarrow P$
T	T	T	T
T	F	F	F
F	T	F	T
F	F	F	T

Que 1 Identify which is well formed formula

1) $P \wedge Q$

→ It is not a well formed formula because in the above equation there is a missing of parenthesis.

2) (P)

→ It is a well-formed formula because extra parenthesis is allowed in well formed formula.

3) $(P \wedge)$

→ It is not well-formed formula because second operand is missing.

4) $((P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P))$

→ it is well-formed formula.

Que: For each of the following statement construct the truth table and identify whether it is tautology, contradiction or contingency.

1) $(P \wedge Q) \vee \neg P$

P	Q	$\neg P$	$P \wedge Q$	$(P \wedge Q) \vee \neg P$
T	T	F	T	T
T	F	F	F	F
F	T	T	F	T
F	F	T	F	T

∴ It is contingency

2) $(P \rightarrow Q) \vee (Q \rightarrow P)$

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \vee (Q \rightarrow P)$
T	T	T	T	T
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

∴ It is tautology.

3) $(P \vee Q) \wedge (\neg P \wedge \neg Q)$

P	Q	$\neg P$	$\neg Q$	$P \vee Q$	$\neg P \wedge \neg Q$	$(P \vee Q) \wedge (\neg P \wedge \neg Q)$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	F

∴ it is contradiction.

4) $\neg(P \vee Q) \leftrightarrow (\neg P \vee \neg Q)$

P	Q	$\neg P$	$\neg Q$	$P \vee Q$	$\neg(P \vee Q)$	$\neg P \vee \neg Q$	$\neg(P \vee Q) \leftrightarrow (\neg P \vee \neg Q)$
T	T	F	F	T	F	F	T
T	F	F	T	T	F	T	F
F	T	T	F	T	F	T	F
F	F	T	T	F	T	T	T

∴ it is contingency.

$$5) (P \rightarrow Q) \wedge (\neg Q \rightarrow \neg P)$$

P	Q	$\neg Q$	$\neg P$	$P \rightarrow Q$	$\neg Q \rightarrow \neg P$	$(P \rightarrow Q) \wedge (\neg Q \rightarrow \neg P)$
T	T	F	F	T	T	T
T	F	T	F	F	F	F
F	T	F	T	T	T	T
F	F	T	T	T	T	T

it is contingency.

* Equivalence Formula.

Let A and B be two statement formulas. and let $P_1, P_2, \dots, P_n$ denote all the variables occurring in both A and B. If the truth value of A is equal to B for everyone of the 2^n possible sets of truth values are assigned, then A and B are said to be equivalent.

(Q:) Construct Truth Table to show them it is equivalent or Not.

$$(1) P \leftrightarrow Q \iff (P \rightarrow Q \wedge Q \rightarrow P)$$

(1)			(2)		
P	Q	$P \leftrightarrow Q$	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q \wedge Q \rightarrow P)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

$$(2) P \wedge (Q \vee R) \iff (P \wedge Q) \wedge R$$

(1)				(2)		
P	Q	R	$Q \vee R$	$P \wedge (Q \vee R)$	$P \wedge Q$	$(P \wedge Q) \wedge R$
T	T	T	T	T	T	T
T	T	F	T	T	T	F
T	F	T	T	T	F	F
T	F	F	F	F	F	F
F	T	T	T	F	F	F
F	T	F	T	F	F	F
F	F	T	T	F	F	F
F	F	F	F	F	F	F

$$(3) P \vee \neg Q \iff \neg(P \wedge Q)$$

(1)			(2)		
P	Q	$\neg Q$	$P \vee \neg Q$	$P \wedge Q$	$\neg(P \wedge Q)$
T	T	F	T	T	F
T	F	T	T	F	T
F	T	F	F	F	T
F	F	T	T	F	T

• Duality Law

Two formulas A and A^* is said to be dual. for each other one can be obtain from other by replacing conjunction by disjunction and disjunction by conjunction.

Eg., a) $(P \vee Q) \wedge R \rightarrow (P \wedge Q) \vee R$
 b) $\neg(P \vee Q) \wedge (P \wedge \neg(Q \vee \neg S)) \rightarrow \neg(P \wedge Q) \vee (P \vee \neg(Q \wedge \neg S))$

* Equivalent formulas.

Disjunction ($\vee$) Conjunction ($\wedge$)

(1) $P \vee P \Leftrightarrow P$ $P \wedge P \Leftrightarrow P$ Idempotent law

(2) $(P \vee Q) \vee R \Leftrightarrow (P \wedge Q) \wedge R \Leftrightarrow$ Associative law
 $R \vee (Q \vee R)$ $P \wedge (Q \wedge R)$

(3) $P \vee Q \Leftrightarrow Q \vee P$ $P \wedge Q \Leftrightarrow Q \wedge P$ Commutative law

(4) $P \vee (Q \wedge R) \Leftrightarrow P \wedge (Q \vee R) \Leftrightarrow$ Distributive law
 $(P \vee Q) \wedge (P \vee R)$ $(P \wedge Q) \vee (P \wedge R)$

(5) $P \vee (P \wedge Q) \Leftrightarrow P$ $P \wedge (P \vee Q) \Leftrightarrow P$ Absorption law

(6) $\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$ $\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$ De Morgan's law

* Rules of Implication

Disjunction ($\vee$) Conjunction ($\wedge$)

(1) $P \vee F \Leftrightarrow P$ $P \wedge T \Leftrightarrow P$

(2) $P \vee T \Leftrightarrow T$ $P \wedge F \Leftrightarrow F$

(3) $T \Leftrightarrow P$ $P \Leftrightarrow T$ $q \Leftrightarrow F$ $F \Leftrightarrow q$

(4) $P \vee \neg P \Leftrightarrow T$ $P \wedge \neg P \Leftrightarrow F$

(5) $P \vee Q \Leftrightarrow Q$ $P \wedge Q \Leftrightarrow P$

(6) $P \Leftrightarrow P \rightarrow Q \Leftrightarrow \neg P \vee Q$

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Que 1) $P \wedge (P \vee Q) \Leftrightarrow P$
 $\rightarrow$ LHS: $P \wedge (P \vee Q)$

$= (P \wedge P) \vee (P \wedge Q)$ -- distributive law

$= P \vee (P \wedge Q)$ -- Idempotent law

$= P$ -- Absorption law

Que 2) $P \vee (\neg P \vee Q) \Leftrightarrow P \vee Q$

$\rightarrow$ LHS: $P \vee (\neg P \vee Q)$

$= (P \vee \neg P) \wedge (P \vee Q)$ -- Distributive law

$P \Leftrightarrow T \rightarrow$ $= T \wedge Q$ -- Implication law

$= P \wedge Q$ -- Implication law

que 3) $(P \wedge Q) \rightarrow r \Leftrightarrow P \rightarrow (Q \rightarrow r)$

→ LHS: $(P \wedge Q) \rightarrow r$
 $= \sim(P \wedge Q) \vee r$ ----- Implication law
 $= (\sim P \vee \sim Q) \vee r$ ----- Demorgan's law
 $= \sim P \vee (\sim Q \vee r)$ ----- Association law
 $= \sim P \vee (Q \rightarrow r)$
 $= P \rightarrow (Q \rightarrow r)$

que 4) Show that $(\sim P \wedge (\sim Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow P$

→ LHS: $(\sim P \wedge (\sim Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R)$
 $= (\sim P \wedge (\sim Q \wedge R)) \vee (Q \vee P) \wedge R$ -- distributive law
 $= (\sim P \wedge \sim Q) \wedge R \vee (Q \vee P) \wedge R$ -- associative law
 $= (\sim(P \vee Q) \wedge R) \vee (Q \vee P) \wedge R$ -- Demorgan's law
 $= (\sim(P \vee Q) \wedge R) \vee (P \vee Q) \wedge R$ -- commutative law
 $= \sim(P \vee Q) \vee (P \vee Q) \wedge R$ -- distributive law
 $= T \wedge R$ -- $\sim P \vee P \Leftrightarrow T$
 $= P$ -- $P \wedge T \Leftrightarrow P$

que 5) Show that $P \rightarrow (Q \rightarrow R) \Leftrightarrow (P \wedge Q) \rightarrow R$

→ LHS: $P \rightarrow (Q \rightarrow R)$
 $= \sim P \vee (Q \rightarrow R)$ -- implication
 $= \sim P \vee (\sim Q \vee R)$ -- implication
 $= (\sim P \vee \sim Q) \vee R$ -- associative law
 $= \sim(P \wedge Q) \vee R$ -- Demorgan's law
 $= (P \wedge Q) \rightarrow R$ -- implication

6) $\sim(P \wedge Q) \rightarrow (\sim P \vee (\sim P \vee Q)) \Leftrightarrow (\sim P \vee Q)$

7) $\sim(P \wedge Q)$

- Well ordering principal

proof: Every non-empty set of natural numbers has a least element.

Proof: let S be a non-empty set of positive integers.

we want to prove that S has a least element, say m such that

least element $m \leq x, \forall x \in S$

Now, consider the set of all positive integers $N = \{1, 2, 3, 4, 5, \dots\}$

start checking from upward

- If $1 \in S$, then clearly 1 is the smallest element S
- If $1 \notin S$, check 2
- If $2 \notin S$, check for 3 and so on.

Since S is non-empty; there must exist some positive integers that belongs to S.

let m be the first (smallest) positive integer that appears in S during this checking process.

By construction, there is no integer smaller than m belongs to S .

Hence m is the smallest element of S .

$\therefore$ Therefore, every non-empty set of positive integer has a least elements.

Ex: $m \leq x, \forall x \in S$

$S = \{3, 6, 8, 10\}$

let's say

$$m = 3$$

$$3 \leq 3$$

$$3 \leq 6$$

$$3 \leq 8$$

$$3 \leq 10$$

$\therefore$ which is true for all.

• Mathematical Induction

Q: Prove by mathematical induction that

$$P(n) = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \text{ for}$$

any $n \geq 1$

Solⁿ: In the basic step we have to prove particular statement is true for $n=1$,

$$P(1) = \frac{n(n+1)}{2}$$

$$= \frac{1(1+1)}{2}$$

$$= \frac{2}{2} = 1$$

which is clearly true

we have to show that it is true

for $P(k)$

where k is $k \geq 1$.

$$P(k) = \frac{k(k+1)}{2} \text{ (assume)}$$

$$P(k) = 1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2} \text{ --- eqⁿ (1)}$$

$$P(k+1) = 1 + 2 + 3 + \dots + k + k+1 = \frac{(k+1)(k+1+1)}{2}$$

The LHS of $P(k+1)$

$$1 + 2 + 3 + \dots + k + k+1$$

$$= \frac{k(k+1)}{2} + k+1 \text{ --- from eq (1)}$$

$$= (k+1) \left(\frac{k+1}{2} \right) \times \frac{2}{2}$$

$$= (k+1) \left(\frac{(k+1) \times 2}{2} \right)$$

$$= \frac{(k+1)(k+2)}{2}$$

$$P(k+1) = \frac{(k+1)(k+1+1)}{2}$$

hence it is true for all n .

2. Q) Show that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

$n \geq 1$ by mathematical induction.

Soln:

In the basic step we have to prove that particular statement is true for $n=1$.

$$P(n) = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$P(1) = \frac{1(1+1)(2(1)+1)}{6}$$

$$= \frac{1 \times 2 \times 3}{6} = \frac{2}{2} = 1$$

$\therefore$ which is clearly true

Assume $P(k)$ is true where $n \geq 1$

$$P(k) = 1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6} \quad \text{--- eqn (1)}$$

$$P(k+1) = 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2$$

$$= \frac{(k+1)(k+1+1)(2(k+1)+1)}{6}$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

LHS of $P(k+1)$

$$= 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2$$

$$= \frac{1}{6} (k+1)(2k+1) + (k+1)^2$$

$$= (k+1) \left[\frac{k(2k+1) + (k+1)}{6} \right]$$

$$= \frac{k+1}{6} [k(2k+1) + (k+1)]$$

$$= \frac{k+1}{6} [2k^2 + k + k + 1]$$

$$= \frac{k+1}{6} [2k^2 + 2k + 1]$$

$$= \frac{k+1}{6} [2k^2 + 4k + 3k + 1]$$

$$= \frac{k+1}{6} [2k(k+2) + 3(k+2)]$$

$$= \frac{k+1}{6} [(k+2)(2k+3)]$$

dividend: divisor quotient
+ remainder
for $a = bq_1 + r_1$

$r_1 = \text{Remainder}$
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* Euclidean algorithm:-

The GCD of two integers a & b can be determined by the process known as Euclidean algorithm.

Proof:- Let a and b both are positive and $a > b$
There exists integers q_1 and r_1

$$a = bq_1 + r_1, \quad 0 \leq r_1 < b \rightarrow (1)$$

Again there exists integer q_2 and r_2

Such that

$$b = r_1q_2 + r_2, \quad 0 \leq r_2 < r_1 \rightarrow (2)$$

Continuing the process we get

$$r_1 = r_2q_3 + r_3, \quad 0 \leq r_3 < r_2 \rightarrow (3)$$

$$r_2 = r_3q_4 + r_4, \quad 0 \leq r_4 < r_3 \rightarrow (4)$$

⋮

$$r_{n-1} = r_nq_n + r_n, \quad r_n = 0 \rightarrow (5)$$

where $q_n \geq 2$

Here we have

$$a > b > r_1 > r_2 \dots$$

Thus these numbers form a decreasing sequence of non-negative integers.

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It follows that $r_n = 0$ for some integer n

The process ends at this stage

The set of equation (1)-(5) is called Euclid's algorithm for GCD(a, b)

Q1) Find the GCD of (12, 33)

Q	A	B	R
2	33	12	9
1	12	9	3
3	9	3	0
	3	0	

The GCD of (12, 33) is 3

Q2) Find GCD of (750, 900)

Q	A	B	R
1	900	750	150
5	750	150	0
	150	0	

The GCD of (750, 900) is 150

Algorithm :-

- ① Divide a by b and find remainder r
 $a = bq + r$
- ② if $r = 0$, then $GCD = b$
- ③ if $r \neq 0$ then replace a by b and b by r
- ④ repeat the steps until remainder = 0
- ⑤ The last non-zero remainder is GCD

③ (252, 105)

Q	A	B	R
2	252	105	42
2	105	42	21
2	42	21	0

$GCD = 21$ of (252, 105)

④ (1005, 105)

Q	A	B	R
9	1005	105	60
1	105	60	45
1	60	45	15
3	45	15	0

$GCD = 15$

⑤ (13, 31)

Q	A	B	R
2	31	13	5
2	13	5	3
1	5	3	2
1	3	2	1
2	2	1	0

$GCD = 1$

⑥ (48, 18)

Q	A	B	R
2	48	18	12
1	18	12	6
2	12	6	0

$GCD = 6$

⑦ (56, 36)

Q	A	B	R
1	56	36	20
1	36	20	16
1	20	16	4
4	16	4	0

$GCD = 4$